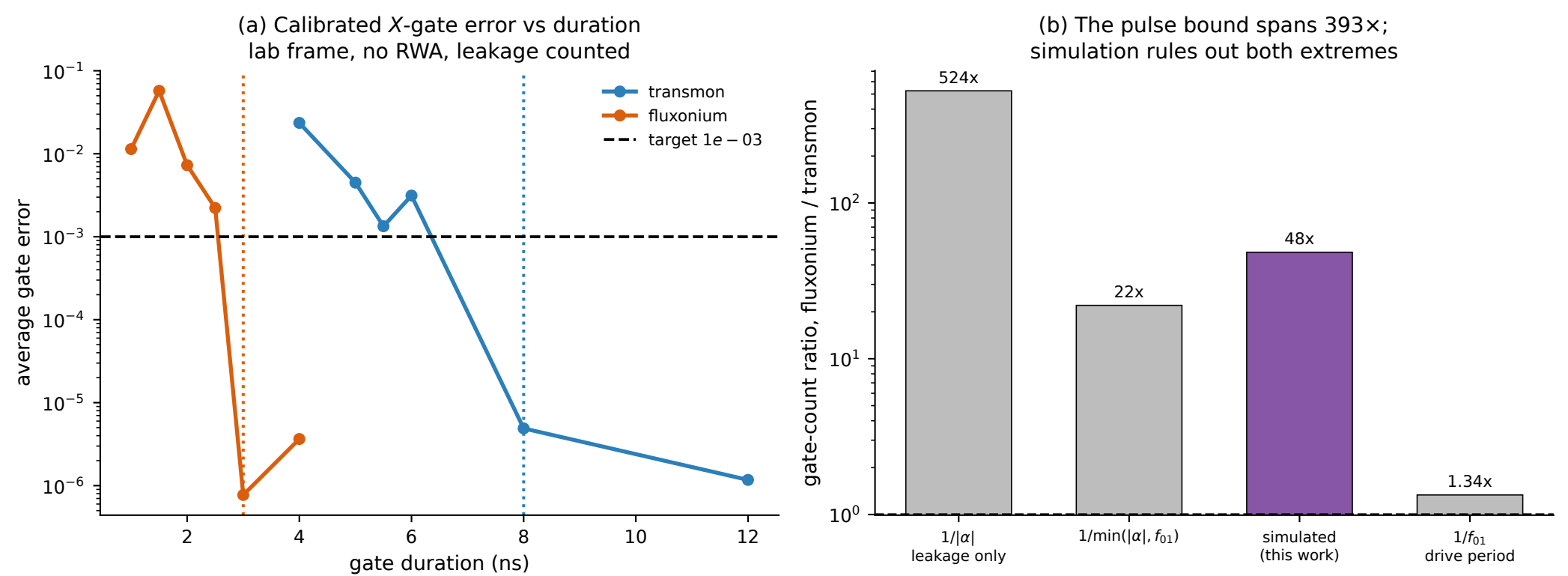

(a) Calibrated X-gate error vs duration
lab frame, no RWA, leakage counted
average gate error
gate duration (ns)
transmon
fluxonium
target 1e − 03
(b) The pulse bound spans 393×;
simulation rules out both extremes
gate-count ratio, fluxonium / transmon
524x
22x
48x
1.34x
1/|α|
leakage only
1/min(|α|, f01)
simulated
(this work)
1/f01
drive period